

CUSTOMER CHURN ANALYSIS

Internship Project Report

A data-driven analysis of customer churn patterns, recurring revenue loss, customer behavior, and retention opportunities.

Total Customers	Churned Customers	Churn Rate
600	247	41.17%
MRR Loss	Average Monthly Charge	Estimated CLV
₹ 240,102.48	₹ 869.03	₹ 2,111.01

Prepared as an internship data analytics project

1. Project Overview

Customer churn refers to customers discontinuing a service. This project analyzes a customer dataset to identify patterns associated with churn and estimate the recurring monthly revenue affected by customer loss.

2. Project Objectives

- Clean and prepare the customer churn dataset for analysis.
- Measure the overall customer churn rate.
- Identify customer groups and behaviors associated with higher churn.
- Estimate recurring monthly revenue loss caused by churned customers.
- Calculate a simplified customer lifetime value estimate.
- Develop practical retention recommendations from observed patterns.

3. Dataset Overview

The cleaned dataset contains 600 customer records and 14 columns. It includes customer demographics, region, tenure, subscription information, charges, contract type, support activity, payment behavior, AutoPay status, churn status, and a derived tenure group.

Category	Examples
Customer information	Customer ID, Age, Region
Subscription	Subscription Type, Contract Type, Tenure

Financial	Monthly Charges, Total Charges
Customer behavior	Support Tickets, Late Payments
Payment	Payment Method, AutoPay
Target variable	Churn
Derived feature	Tenure Group

4. Data Cleaning and Preprocessing

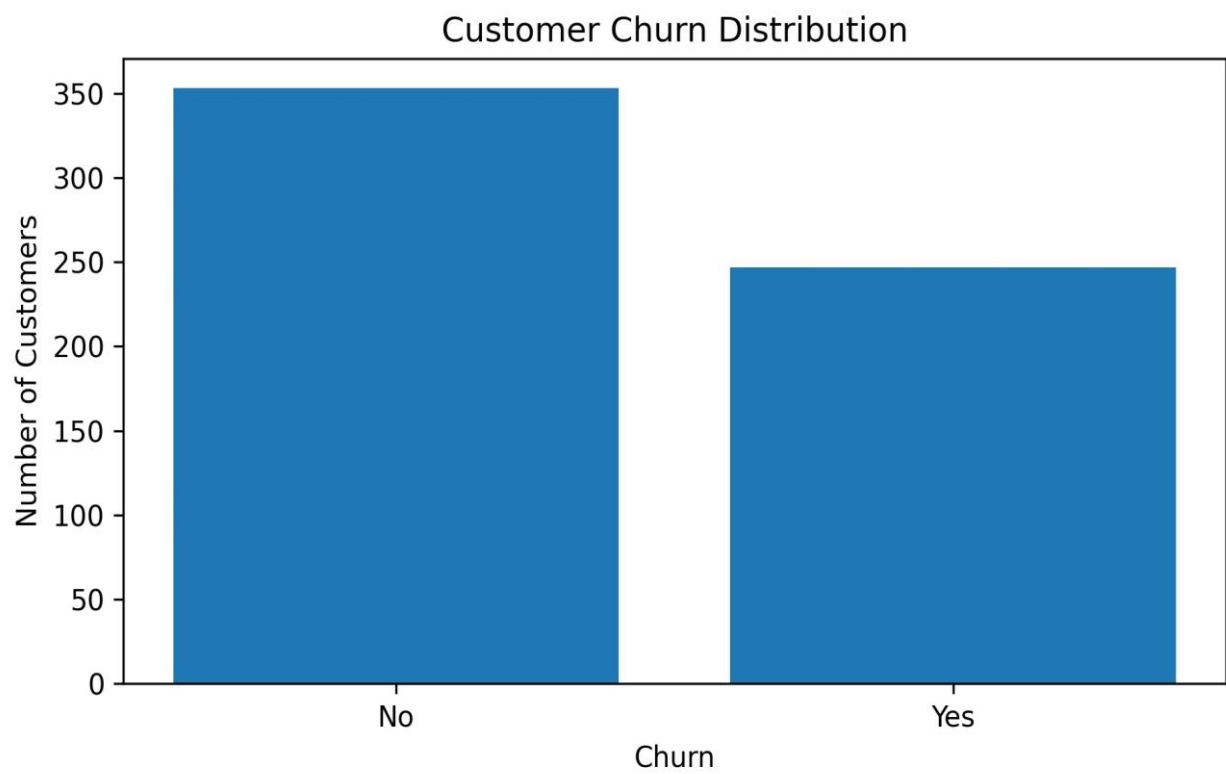
- Loaded the raw customer churn CSV dataset using pandas.
- Checked the dataset structure, columns, data types, and missing values.
- Handled and verified missing values in important fields.
- Created the TenureGroup feature using tenure-month ranges.
- Verified the cleaned dataset contains 600 rows and 14 columns.
- Verified the final cleaned dataset contains no missing values in the analyzed fields.

5. Key Metrics

Metric	Result
Total Customers	600
Churned Customers	247
Churn Rate	41.17%
MRR Loss from Churned Customers	₹ 240,102.48
Average Monthly Charge	₹ 869.03
Estimated Customer Lifetime	2.43 months
Estimated CLV	₹ 2,111.01

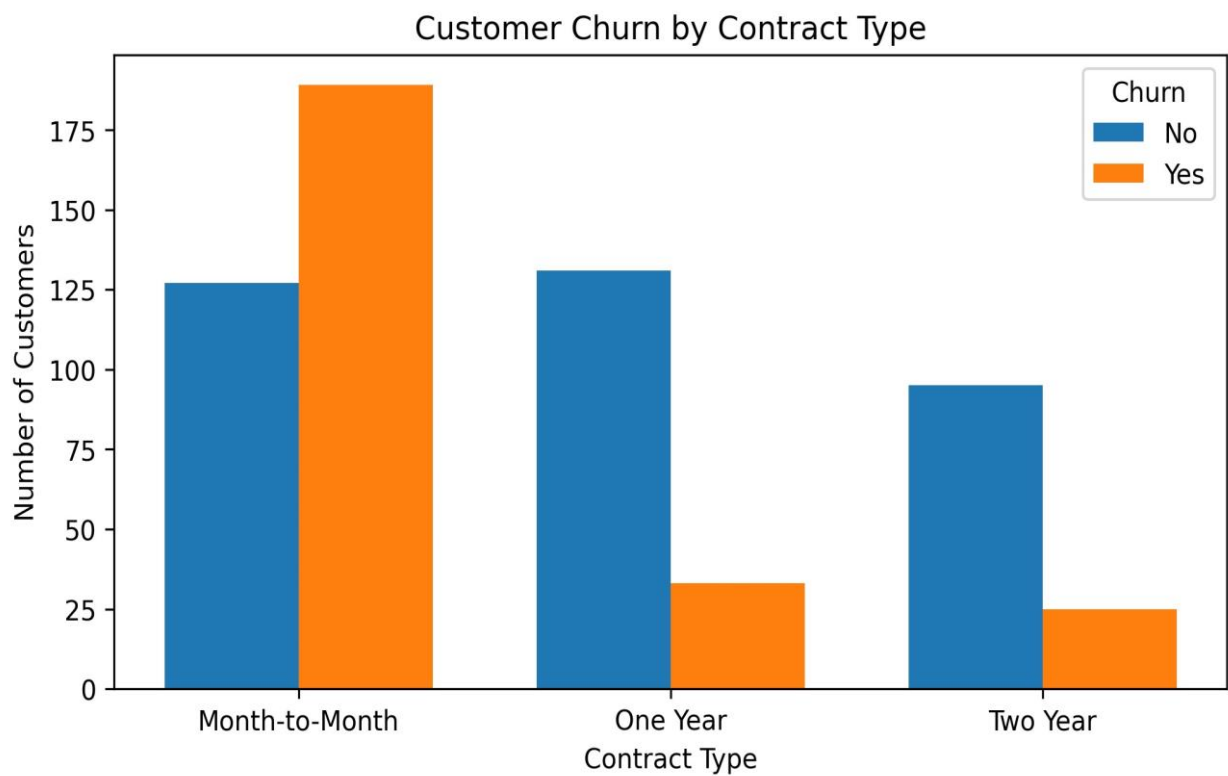
CLV note: This is a simplified project estimate calculated as average monthly charge multiplied by the reciprocal of the observed churn rate. It is not a production-grade CLV model.

Customer Churn Distribution



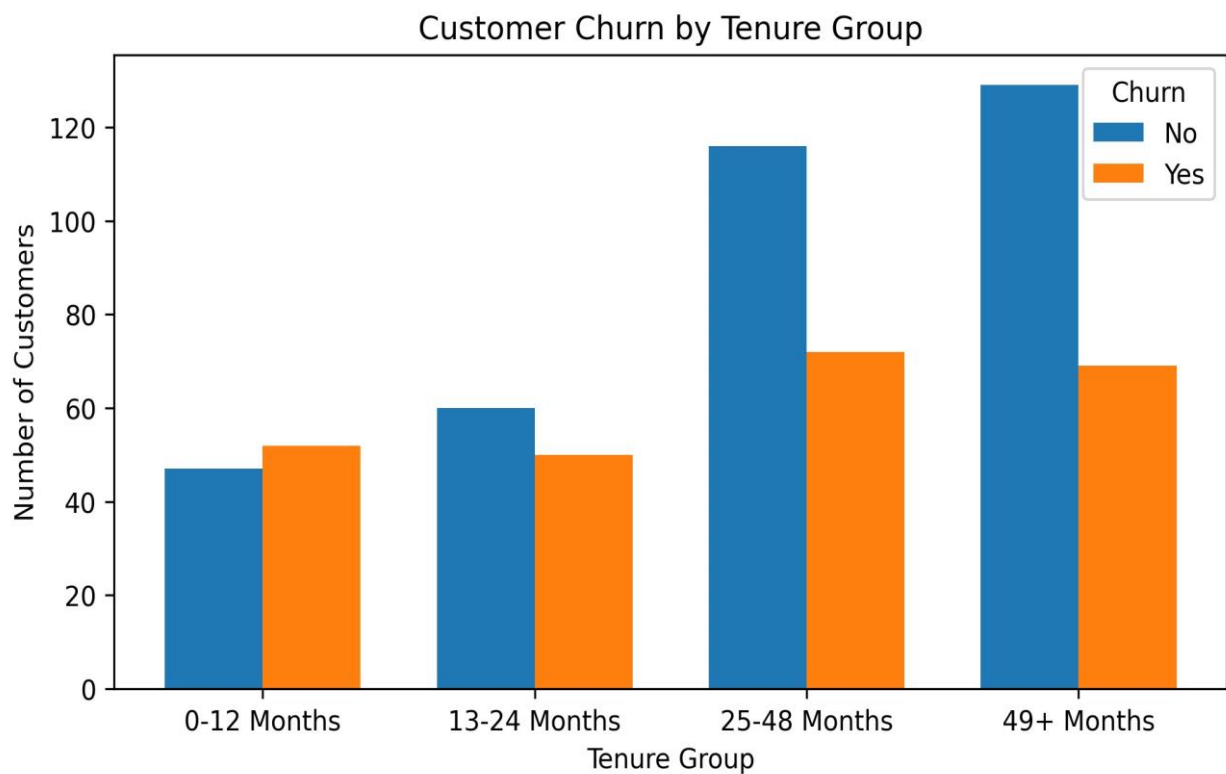
247 of 600 customers are classified as churned, producing an overall churn rate of 41.17%.

Customer Churn by Contract Type



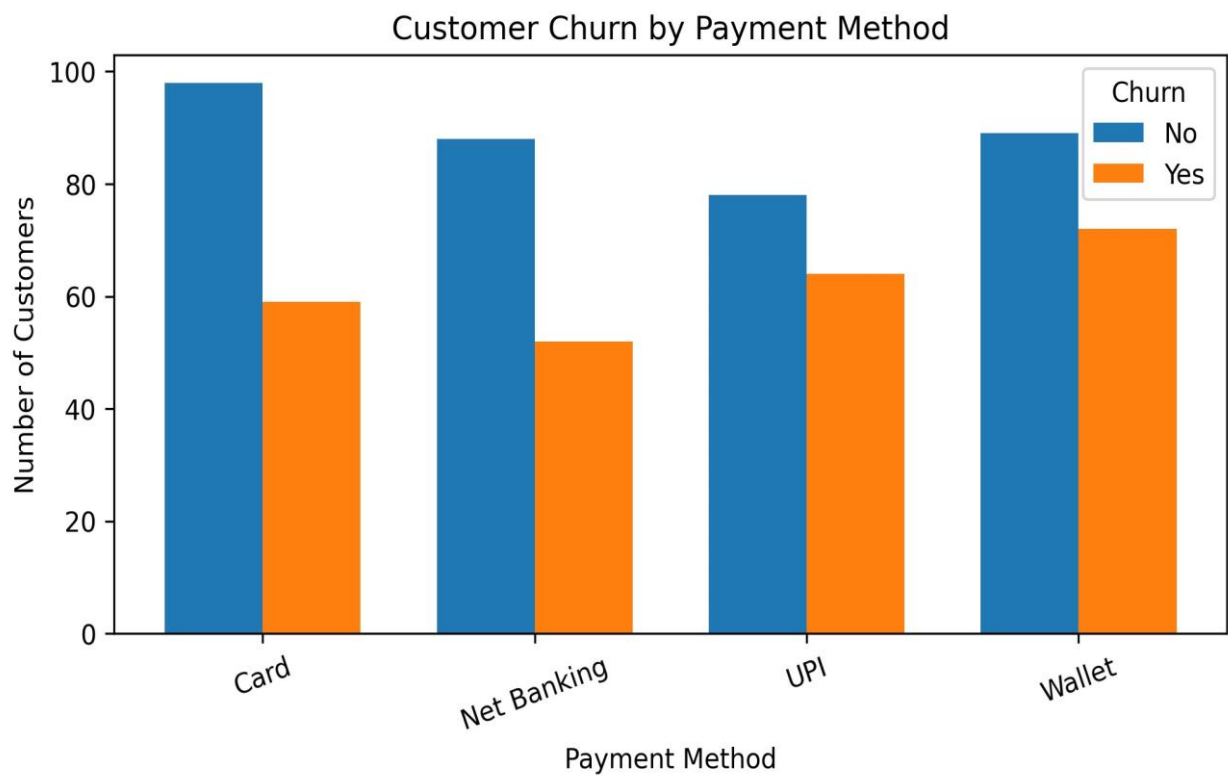
Month-to-month customers have an observed churn rate of 59.81%, compared with about 20.12% for One Year and 20.83% for Two Year contracts.

Customer Churn by Tenure Group



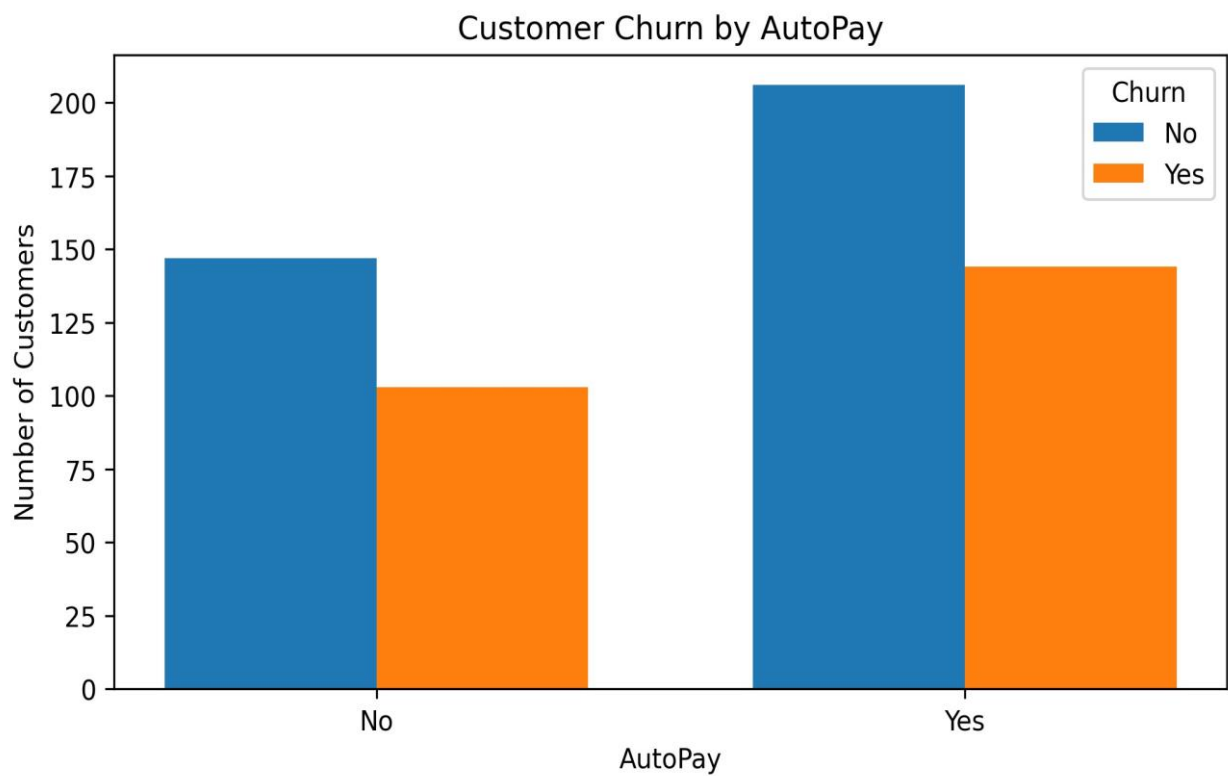
The 0–12 month group has the highest observed churn rate at 52.53%. Longer-tenure groups show lower observed rates.

Customer Churn by Payment Method



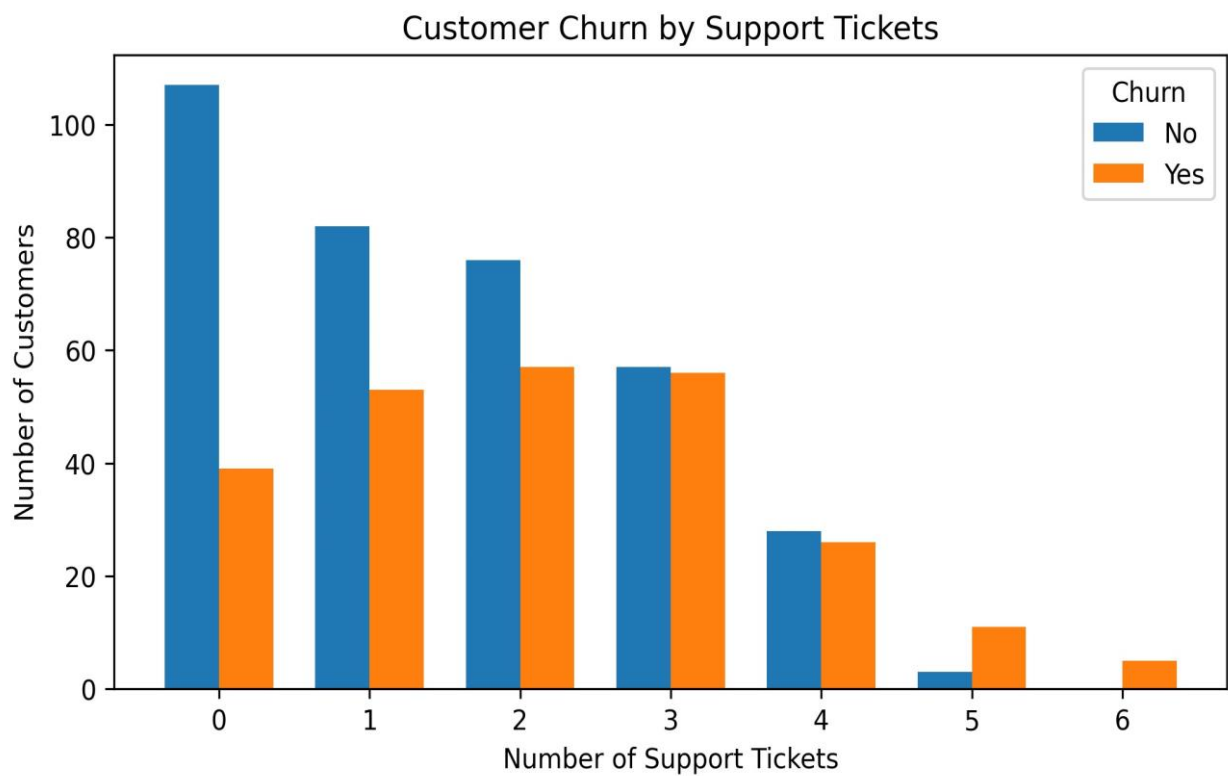
Churn differs across payment methods in this sample. These differences are associations and do not by themselves establish causation.

Customer Churn by AutoPay



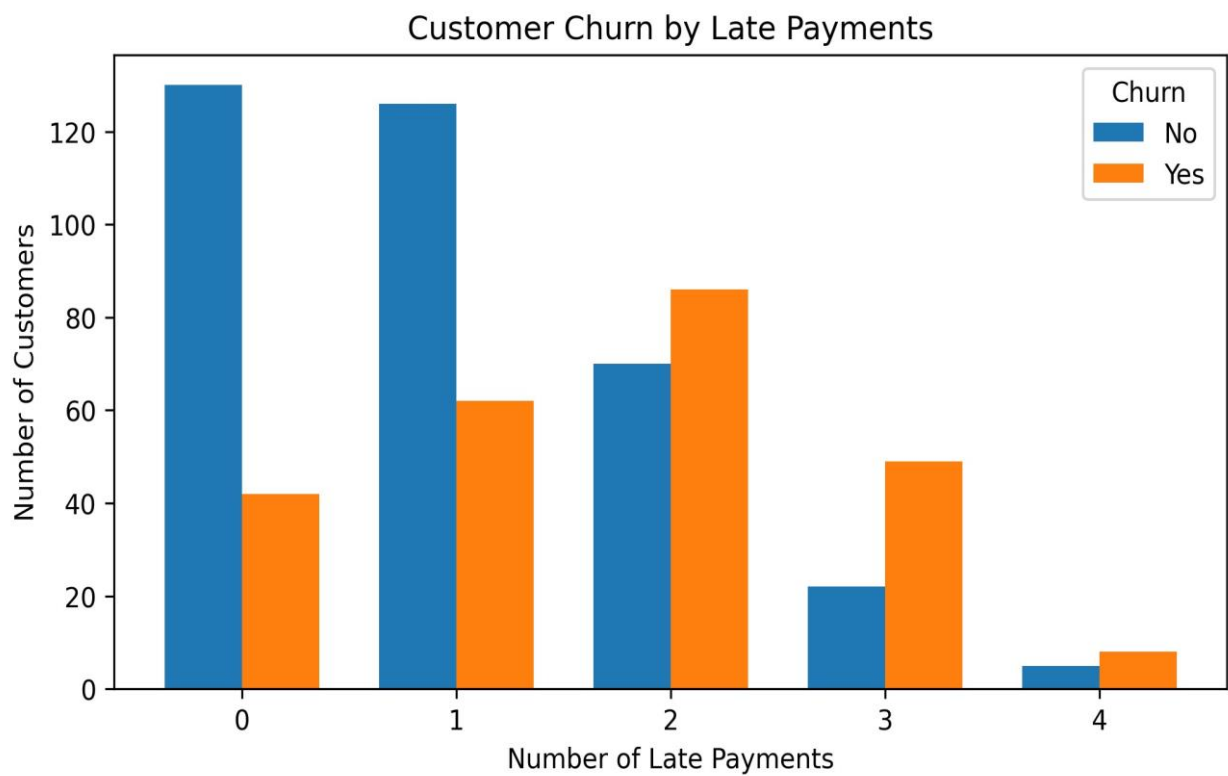
The AutoPay groups have very similar observed churn rates in this dataset.

Customer Churn by Support Tickets



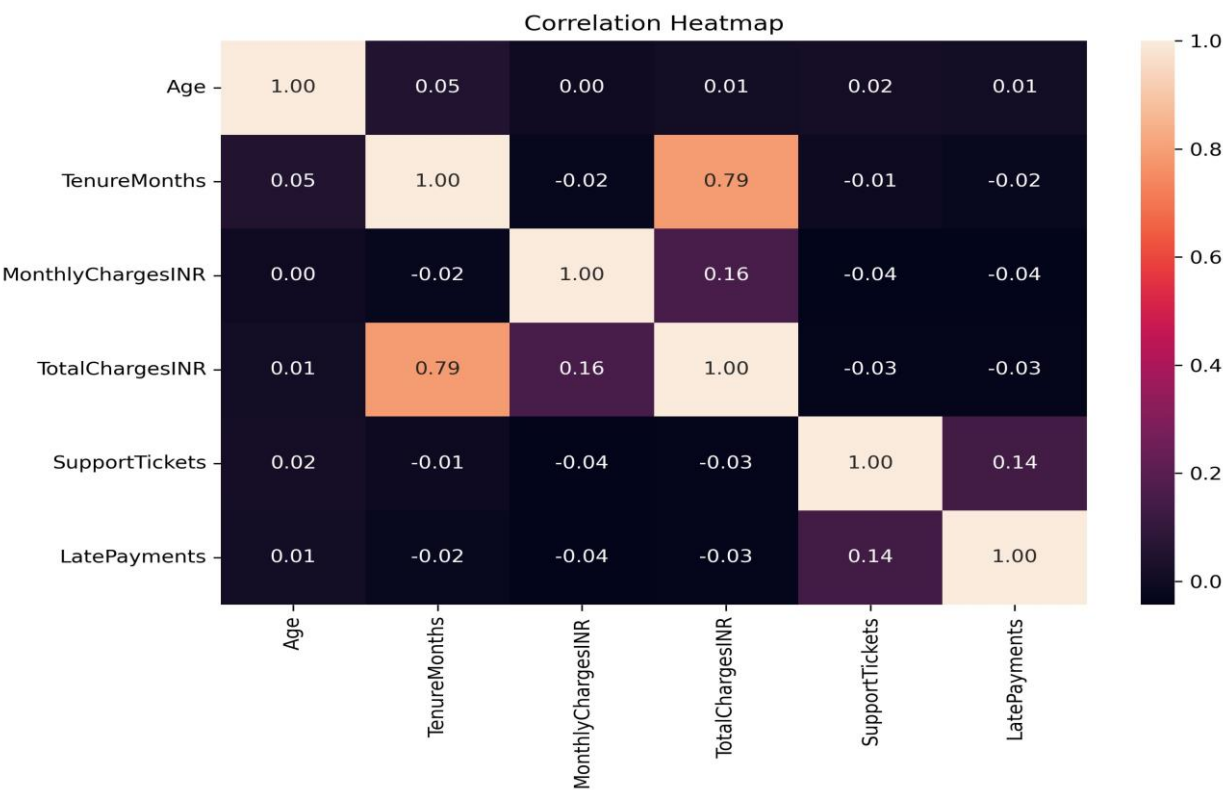
Higher support-ticket counts are generally associated with higher observed churn in this dataset.

Customer Churn by Late Payments



Higher late-payment counts are associated with higher observed churn across several groups.

Correlation Heatmap



The correlation heatmap summarizes relationships among numerical variables. Correlation indicates association, not causation.

14. Key Findings

- The overall churn rate is 41.17%, with 247 churned customers.
- Churned customers represent approximately ₹ 240,102.48 in monthly recurring charges in the analyzed dataset.
- Month-to-month contracts have an observed churn rate of 59.81%, higher than the longer-term contract groups.
- The 0–12 month tenure group has an observed churn rate of 52.53%.
- Payment method and AutoPay status show differences in churn patterns, although the differences should be interpreted as associations.
- Higher support-ticket and late-payment levels are associated with higher churn in several groups.
- The correlation analysis identifies numerical relationships but does not establish causation.

15. Business Recommendations

- Strengthen onboarding and early-stage customer engagement.
- Encourage suitable customers to consider longer-term plans through transparent value-based offers.
- Use timely payment reminders and improve the payment experience.
- Monitor customers with repeated support tickets and provide proactive assistance.
- Use tenure, contract, payment, support, and activity signals for churn-risk monitoring.
- Track churn rate and recurring revenue loss regularly to evaluate retention actions.

16. Conclusion

This customer churn analysis demonstrates how data cleaning, exploratory analysis, visualization, and business metrics can be combined to understand customer retention patterns. The results highlight measurable differences across contract type, tenure, payment behavior, support activity, and late payments. These insights can support targeted retention monitoring and help organizations focus customer-success efforts on patterns associated with higher churn.

17. Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- CSV
- ReportLab